



SYSTEM: 7ep

system: "7ep"

services:

- name: "Authentication Service"

responsibilities:

- "Register librarians with username and password"
- "Enforce unique librarian usernames"
- "Validate required credentials and password strength rules"
- "Authenticate librarians and securely validate credentials"
- "Manage and validate librarian sessions"
- "Handle access denial responses securely"

dependencies:

- "Audit Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Book Catalog Service"

responsibilities:

- "Register books in the catalog by title"
- "Prevent duplicate book registration"
- "Delete registered books"
- "Coordinate loan link removal when deleting loaned books"
- "List registered books"
- "Consult books by identifier"
- "Consult books by title"
- "List books available for loan"
- "Handle empty catalog and search result notifications"

dependencies:

- "Audit Service"
- "Loan Management Service"

suggested_endpoints:

- "/api/book-catalog-service"

- name: "Reader Management Service"

responsibilities:

- "Register readers eligible for loans"
- "Prevent duplicate reader registration"
- "Delete registered readers"
- "Coordinate loan link removal when deleting readers with borrowed books"
- "List registered readers"
- "Consult readers by identifier"

- "Consult readers by name"
- "Validate search parameters to prevent simultaneous identifier and name queries"
- "Handle empty reader and search result notifications"

dependencies:

- "Audit Service"
- "Loan Management Service"

suggested_endpoints:

- "/api/reader-management-service"

- name: "Loan Management Service"

responsibilities:

- "Register book loans with loan date"
- "Validate reader eligibility"
- "Validate book existence and availability"
- "Prevent loans for already borrowed books"
- "Ensure single-reader book allocation"
- "Allow multiple concurrent loans per reader"
- "Maintain loan links and consistency across books readers and loans"
- "Provide feedback when loans cannot be completed"

dependencies:

- "Audit Service"
- "Book Catalog Service"
- "Reader Management Service"

suggested_endpoints:

- "/api/loan-management-service"

- name: "Environment Management Service"

responsibilities:

- "Clear data and base structure to reset environment"
- "Recreate base structure"
- "Restore clean environment state ready for tests"
- "Coordinate involved functional areas for consistent data restoration"

dependencies:

- "Authentication Service"
- "Book Catalog Service"
- "Loan Management Service"
- "Reader Management Service"

suggested_endpoints:

- "/api/environment-management-service"

- name: "Demo Utility Service"

responsibilities:

- "Calculate sum of two integers"
- "Calculate Fibonacci sequence values using multiple approaches"
- "Calculate Ackermann function results using multiple approaches"
- "Validate input formats and report calculation errors"

- "Provide service catalog for testing endpoints"

dependencies:

- ""

suggested_endpoints:

- "/api/demo-utility-service"

- name: "Audit Service"

responsibilities:

- "Record relevant operations from registration authentication consultation and loan flows"
- "Maintain centralized event recording decoupled from functional domains"
- "Provide failure and behavior tracking for traceability and problem analysis"

dependencies:

- ""

suggested_endpoints:

- "/api/audit-service"

interactions:

- from: "Authentication Service"
to: "Audit Service"
- from: "Book Catalog Service"
to: "Audit Service"
- from: "Book Catalog Service"
to: "Loan Management Service"
- from: "Reader Management Service"
to: "Audit Service"
- from: "Reader Management Service"
to: "Loan Management Service"
- from: "Loan Management Service"
to: "Audit Service"
- from: "Loan Management Service"
to: "Book Catalog Service"
- from: "Loan Management Service"
to: "Reader Management Service"
- from: "Environment Management Service"
to: "Authentication Service"
- from: "Environment Management Service"
to: "Book Catalog Service"
- from: "Environment Management Service"
to: "Loan Management Service"
- from: "Environment Management Service"
to: "Reader Management Service"

SYSTEM: acmeair

system: "acmeair"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate customer credentials"
- "Create and manage customer sessions"
- "Validate active sessions"
- "Invalidate sessions on logout"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Flight Search Service"

responsibilities:

- "Search one-way and round-trip flights"
- "Present flight options with schedules and reference fares"
- "Manage registered airports and routes"

dependencies:

- ""

suggested_endpoints:

- "/api/flight-search-service"

- name: "Booking Service"

responsibilities:

- "Create bookings for one-way and round-trip flights"
- "Generate booking identifiers and record booking dates"
- "Retrieve customer bookings"
- "Cancel bookings and update booking status"

dependencies:

- "Authentication Service"
- "Flight Search Service"

suggested_endpoints:

- "/api/booking-service"

- name: "Customer Profile Service"

responsibilities:

- "Retrieve customer contact and address details"
- "Update contact and address data"
- "Verify customer credentials for profile modifications"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-profile-service"

- name: "Loyalty Service"

responsibilities:

- "Present customer loyalty program status"
- "Present annual and total accumulated miles"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/loyalty-service"

- name: "Environment Administration Service"

responsibilities:

- "Load and reload sample operational data"
- "Reset database and operational records"
- "Provide operational summaries and metrics"

dependencies:

- "Authentication Service"
- "Flight Search Service"
- "Booking Service"
- "Customer Profile Service"

suggested_endpoints:

- "/api/environment-administration-service"

interactions:

- from: "Booking Service"
to: "Authentication Service"
- from: "Booking Service"
to: "Flight Search Service"
- from: "Customer Profile Service"
to: "Authentication Service"
- from: "Loyalty Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "Flight Search Service"
- from: "Environment Administration Service"
to: "Booking Service"
- from: "Environment Administration Service"
to: "Customer Profile Service"

SYSTEM: Cargo-Tracker

system: "Cargo-Tracker"

services:

- name: "Cargo Management Service"

responsibilities:

- "Register new cargo with origin destination and arrival deadline"
- "Assign unique tracking identifier to each cargo"
- "Enforce cargo lifecycle rules and validate registered locations"
- "Update cargo destination and maintain routing status"

dependencies:

- ""

suggested_endpoints:

- "/api/cargo-management-service"

- name: "Route Planning Service"

responsibilities:

- "Select and validate candidate routes based on cargo specifications"
- "Display route legs including load points unload points voyages and expected dates"
- "Reevaluate and replan routes when cargo is off route or destination changes"

dependencies:

- "Cargo Management Service"

suggested_endpoints:

- "/api/route-planning-service"

- name: "Logistics Event Registration Service"

responsibilities:

- "Receive and process logistics events including reception load unload customs and pickup"
- "Validate event references against existing cargo location and voyage"
- "Enforce voyage requirements for load and unload events"
- "Record valid events and isolate rejected registration attempts"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"

suggested_endpoints:

- "/api/logistics-event-registration-service"

- name: "Cargo Tracking Service"

responsibilities:

- "Consolidate operational status and reconstruct cargo history"
- "Provide cargo consultation by tracking identifier with last known location and estimated arrival"
- "Recalculate transport status next expected step and plan adherence"
- "Detect and signal off-route deviations arrival and delivery completion"
- "Provide geographic tracking visualization across cargo journey"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"
- "Logistics Event Registration Service"

suggested_endpoints:

- "/api/cargo-tracking-service"

- name: "Integration Service"

responsibilities:

- "Receive logistics events across multiple operational channels and periodic file inputs"
- "Isolate invalid input records for operational handling"
- "Provide consolidated cargo view and standardized data for external and monitoring channels"
- "Maintain tracking model consistency across integration channels"

dependencies:

- "Cargo Tracking Service"
- "Logistics Event Registration Service"

suggested_endpoints:

- "/api/integration-service"

interactions:

- from: "Route Planning Service"
to: "Cargo Management Service"
- from: "Logistics Event Registration Service"
to: "Cargo Management Service"
- from: "Logistics Event Registration Service"
to: "Route Planning Service"
- from: "Cargo Tracking Service"
to: "Cargo Management Service"
- from: "Cargo Tracking Service"
to: "Route Planning Service"
- from: "Cargo Tracking Service"
to: "Logistics Event Registration Service"
- from: "Integration Service"
to: "Cargo Tracking Service"
- from: "Integration Service"
to: "Logistics Event Registration Service"

SYSTEM: DayTrader7

system: "DayTrader7"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate investors using username and password"
- "Manage active investor sessions"
- "End investor sessions through logout"
- "Enforce access control for unauthenticated requests"

dependencies:

- "Account Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Account Service"

responsibilities:

- "Register new investors"
- "Maintain investor account profiles and financial balances"
- "Validate registration and profile update constraints"
- "Update investor registration and financial data"
- "Track account access statistics and balance adjustments"

dependencies:

- ""

suggested_endpoints:

- "/api/account-service"

- name: "Market Data Service"

responsibilities:

- "Provide single and multiple asset quotes"
- "Calculate and display consolidated market summary"
- "Update traded volume and market indicators following operations"
- "Handle invalid asset queries without navigation interruption"

dependencies:

- ""

suggested_endpoints:

- "/api/market-data-service"

- name: "Trading Service"

responsibilities:

- "Create and track buy and sell orders"
- "Apply operation fees to trading orders"
- "Manage order lifecycle across states"
- "Execute orders immediately or deferred based on operational configuration"
- "Maintain investor order history and completed order alerts"

dependencies:

- "Account Service"
- "Market Data Service"
- "Portfolio Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/trading-service"

- name: "Portfolio Service"

responsibilities:

- "Maintain investor portfolio positions"
- "Register new positions from completed buy orders"
- "Remove positions for completed sell orders"
- "Calculate invested amount and current market value and gain or loss"
- "Provide consolidated financial position summary"
- "Initiate sales of positions directly from portfolio"

dependencies:

- "Account Service"
- "Market Data Service"
- "Trading Service"

suggested_endpoints:

- "/api/portfolio-service"

- name: "Simulation Administration Service"

responsibilities:

- "Manage simulation operational parameters"
- "Configure order processing modes and alert settings and simulation limits"
- "Prepare and rebuild simulation database"
- "Reset operational simulation data"
- "Provide summarized execution statistics"

dependencies:

- "Account Service"
- "Market Data Service"
- "Portfolio Service"
- "Trading Service"

suggested_endpoints:

- "/api/simulation-administration-service"

interactions:

- from: "Authentication Service"
to: "Account Service"
- from: "Trading Service"
to: "Account Service"
- from: "Trading Service"
to: "Market Data Service"
- from: "Trading Service"
to: "Portfolio Service"
- from: "Trading Service"
to: "Simulation Administration Service"
- from: "Portfolio Service"
to: "Account Service"
- from: "Portfolio Service"
to: "Market Data Service"
- from: "Portfolio Service"
to: "Trading Service"
- from: "Simulation Administration Service"
to: "Account Service"
- from: "Simulation Administration Service"
to: "Market Data Service"
- from: "Simulation Administration Service"
to: "Portfolio Service"
- from: "Simulation Administration Service"
to: "Trading Service"

SYSTEM: Jokul

system: "Jokul"

services:

- name: "Authentication Service"

responsibilities:

- "Register users with unique usernames"
- "Authenticate user credentials"
- "Manage user sessions and logout"
- "Control user profiles and access permissions"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Catalog Service"

responsibilities:

- "Display paginated movie listings and total movie count"
- "Display movie details and full cast"
- "Manage categories and filter movies by category"
- "Register new movies with unique titles"
- "Delete movies individually or in batch"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Media Service"

responsibilities:

- "Upload large video files"
- "Manage video storage in local repository and delivery area"
- "Serve video streams for playback"
- "Provide movie download links"
- "Verify video file availability for registered movies"

dependencies:

- "Authentication Service"
- "Catalog Service"

suggested_endpoints:

- "/api/media-service"

interactions:

- from: "Catalog Service"
to: "Authentication Service"
- from: "Media Service"
to: "Authentication Service"
- from: "Media Service"
to: "Catalog Service"

SYSTEM: JpetStore

system: "JpetStore"

services:

- name: "Catalog Service"

responsibilities:

- "Browse virtual store and main catalog"
- "Organize products by animal categories"
- "View product lists and available item details"
- "Search products by keyword"
- "Display item price and stock status"
- "Validate keyword search input"

dependencies:

- "Inventory Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Customer Account Service"

responsibilities:

- "Register new customer identification and access credentials"
- "Manage customer registration data and profile"
- "Configure customer preferences including language and favorite category"
- "Enable personalized product lists and promotional banners based on preferences"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-account-service"

- name: "Authentication Service"

responsibilities:

- "Authenticate customer username and password"
- "Notify invalid login credentials"
- "Manage customer user sessions and logout"
- "Provide access control for protected features"

dependencies:

- "Customer Account Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Shopping Cart Service"

responsibilities:

- "Add and consolidate items in shopping cart"
- "Update item quantities and remove items"
- "Display cart items with unit prices and calculated subtotal"
- "Inform stock availability for cart items"
- "Clear cart after order completion"

dependencies:

- "Catalog Service"
- "Inventory Service"

suggested_endpoints:

- "/api/shopping-cart-service"

- name: "Inventory Service"

responsibilities:

- "Manage item stock availability"
- "Indicate backorder status when items are out of stock"
- "Consult updated stock status during the purchase journey"
- "Register stock decrease upon order completion"

dependencies:

- "Catalog Service"

suggested_endpoints:

- "/api/inventory-service"

- name: "Order Service"

responsibilities:

- "Manage checkout workflow"
- "Compose order from customer account and shopping cart data"
- "Manage payment details and card type selection"
- "Manage separate delivery and billing addresses"
- "Require explicit order confirmation and generate unique order identifier"
- "Record order items and initial order status"
- "Consult completed order list and order details"
- "Restrict order access to authenticated owner"

dependencies:

- "Authentication Service"
- "Customer Account Service"
- "Shopping Cart Service"
- "Inventory Service"

suggested_endpoints:

- "/api/order-service"

interactions:

- from: "Catalog Service"
to: "Inventory Service"
- from: "Customer Account Service"
to: "Authentication Service"
- from: "Authentication Service"
to: "Customer Account Service"
- from: "Shopping Cart Service"
to: "Catalog Service"
- from: "Shopping Cart Service"
to: "Inventory Service"
- from: "Inventory Service"
to: "Catalog Service"
- from: "Order Service"
to: "Authentication Service"

- from: "Order Service"
to: "Customer Account Service"
- from: "Order Service"
to: "Shopping Cart Service"
- from: "Order Service"
to: "Inventory Service"

SYSTEM: Pet-Clinic

system: "Pet-Clinic"

services:

- name: "Owner Service"

responsibilities:

- "Register owner data"
- "Update owner data"
- "Search owners by last name"
- "Retrieve detailed owner information"

dependencies:

- ""

suggested_endpoints:

- "/api/owner-service"

- name: "Animal Service"

responsibilities:

- "Register animals linked to owners"
- "Update animal data"
- "Provide predefined animal types"
- "Validate animal birth date and name uniqueness per owner"

dependencies:

- "Owner Service"

suggested_endpoints:

- "/api/animal-service"

- name: "Visit Service"

responsibilities:

- "Register visits for animals"
- "Maintain visit and service history"
- "Associate visits with animals and owners"
- "Resume pending internal processes"

dependencies:

- "Animal Service"
- "Veterinarian Service"

suggested_endpoints:

- "/api/visit-service"

- name: "Veterinarian Service"

responsibilities:

- "List veterinarians and specialties"
- "Calculate veterinarian workload"
- "Inform next expected visits"
- "Assign visits to veterinarians based on workload"
- "Provide structured veterinarian data"

dependencies:

- "Visit Service"

suggested_endpoints:

- "/api/veterinarian-service"

interactions:

- from: "Animal Service"
to: "Owner Service"
- from: "Visit Service"
to: "Animal Service"
- from: "Visit Service"
to: "Veterinarian Service"
- from: "Veterinarian Service"
to: "Visit Service"

SYSTEM: TNTConcept

system: "TNTConcept"

services:

- name: "Administration and Configuration Service"

responsibilities:

- "Maintain support records"
- "Manage individual user configurations"
- "Provide domain configuration parameters"

dependencies:

- "Identity and Access Service"
- "Workforce Management Service"
- "CRM and Proposals Service"
- "Billing Service"
- "Notification Service"
- "File Storage Service"

suggested_endpoints:

- "/api/administration-and-configuration-service"

- name: "Identity and Access Service"

responsibilities:

- "Register and maintain user data"
- "Manage password changes and resets"
- "Control access by profile and scope"
- "Provide user validation"

dependencies:

- "Administration and Configuration Service"
- "Workforce Management Service"
- "Project and Work Tracking Service"
- "Content and Document Management Service"

suggested_endpoints:

- "/api/identity-and-access-service"

- name: "Workforce Management Service"

responsibilities:

- "Maintain work agreements"
- "Manage vacation requests and balances"
- "Track vacation approvals and contract expirations"

dependencies:

- "Administration and Configuration Service"
- "Identity and Access Service"
- "Notification Service"
- "Project and Work Tracking Service"
- "Commissioning Service"

suggested_endpoints:

- "/api/workforce-management-service"

- name: "Project and Work Tracking Service"

responsibilities:

- "Maintain projects linked to clients"
- "Register internal and external activities"
- "Classify activities and calculate worked hours"
- "Support evidence policies"

dependencies:

- "Administration and Configuration Service"
- "Identity and Access Service"
- "Workforce Management Service"
- "File Storage Service"
- "CRM and Proposals Service"
- "Financial Management Service"
- "Billing Service"
- "Commissioning Service"
- "Notification Service"

suggested_endpoints:

- "/api/project-and-work-tracking-service"

- name: "CRM and Proposals Service"

responsibilities:

- "Maintain organizations and contacts"
- "Register interactions"
- "Manage proposals and proposal lifecycles"

dependencies:

- "Administration and Configuration Service"
- "Project and Work Tracking Service"
- "Billing Service"

suggested_endpoints:

- "/api/crm-and-proposals-service"

- name: "Financial Management Service"

responsibilities:

- "Maintain accounts and financial entries"
- "Track financial indicators"
- "Import billable hours and costs"

dependencies:

- "Billing Service"
- "Project and Work Tracking Service"

suggested_endpoints:

- "/api/financial-management-service"

- name: "Billing Service"

responsibilities:

- "Detail invoices and calculate totals and taxes"
- "Convert proposals to invoices"
- "Manage invoice attachments"

dependencies:

- "Administration and Configuration Service"
- "CRM and Proposals Service"
- "Project and Work Tracking Service"
- "Financial Management Service"
- "File Storage Service"

suggested_endpoints:

- "/api/billing-service"

- name: "Content and Document Management Service"

responsibilities:

- "Organize documents by categories and versions"
- "Associate documents with users and areas"

dependencies:

- "Identity and Access Service"
- "File Storage Service"

suggested_endpoints:

- "/api/content-and-document-management-service"

- name: "Commissioning Service"

responsibilities:

- "Register and manage orders and commissions"
- "Track commission execution"

dependencies:

- "Project and Work Tracking Service"

- "Workforce Management Service"

- "File Storage Service"

- "Notification Service"

suggested_endpoints:

- "/api/commissioning-service"

- name: "Reporting Service"

responsibilities:

- "Consolidate cross-domain business data"

- "Generate and export business reports"

dependencies:

- "Workforce Management Service"

- "Project and Work Tracking Service"

- "CRM and Proposals Service"

- "Financial Management Service"

- "Billing Service"

- "Commissioning Service"

suggested_endpoints:

- "/api/reporting-service"

- name: "Notification Service"

responsibilities:

- "Send email notifications"

- "Process notification rules and settings"

dependencies:

- "Administration and Configuration Service"

- "Workforce Management Service"

- "Project and Work Tracking Service"

- "Commissioning Service"

suggested_endpoints:

- "/api/notification-service"

- name: "File Storage Service"

responsibilities:

- "Store and retrieve files"

- "Manage file storage policies and settings"

dependencies:

- "Administration and Configuration Service"

- "Project and Work Tracking Service"

- "Billing Service"

- "Content and Document Management Service"

- "Commissioning Service"

suggested_endpoints:

- "/api/file-storage-service"

interactions:

- from: "Administration and Configuration Service"

to: "Identity and Access Service"

- from: "Administration and Configuration Service"

to: "Workforce Management Service"

- from: "Administration and Configuration Service"

to: "CRM and Proposals Service"

- from: "Administration and Configuration Service"

to: "Billing Service"

- from: "Administration and Configuration Service"

to: "Notification Service"

- from: "Administration and Configuration Service"

to: "File Storage Service"

- from: "Identity and Access Service"

to: "Administration and Configuration Service"

- from: "Identity and Access Service"

to: "Workforce Management Service"

- from: "Identity and Access Service"

to: "Project and Work Tracking Service"

- from: "Identity and Access Service"

to: "Content and Document Management Service"

- from: "Workforce Management Service"

to: "Administration and Configuration Service"

- from: "Workforce Management Service"

to: "Identity and Access Service"

- from: "Workforce Management Service"

to: "Notification Service"

- from: "Workforce Management Service"

to: "Project and Work Tracking Service"

- from: "Workforce Management Service"

to: "Commissioning Service"

- from: "Project and Work Tracking Service"

to: "Administration and Configuration Service"

- from: "Project and Work Tracking Service"

to: "Identity and Access Service"

- from: "Project and Work Tracking Service"

to: "Workforce Management Service"

- from: "Project and Work Tracking Service"

to: "File Storage Service"

- from: "Project and Work Tracking Service"

to: "CRM and Proposals Service"

- from: "Project and Work Tracking Service"

to: "Financial Management Service"

- from: "Project and Work Tracking Service"

to: "Billing Service"

- from: "Project and Work Tracking Service"
to: "Commissioning Service"

- from: "Project and Work Tracking Service"
to: "Notification Service"

- from: "CRM and Proposals Service"
to: "Administration and Configuration Service"

- from: "CRM and Proposals Service"
to: "Project and Work Tracking Service"

- from: "CRM and Proposals Service"
to: "Billing Service"

- from: "Financial Management Service"
to: "Billing Service"

- from: "Financial Management Service"
to: "Project and Work Tracking Service"

- from: "Billing Service"
to: "Administration and Configuration Service"

- from: "Billing Service"
to: "CRM and Proposals Service"

- from: "Billing Service"
to: "Project and Work Tracking Service"

- from: "Billing Service"
to: "Financial Management Service"

- from: "Billing Service"
to: "File Storage Service"

- from: "Content and Document Management Service"
to: "Identity and Access Service"

- from: "Content and Document Management Service"
to: "File Storage Service"

- from: "Commissioning Service"
to: "Project and Work Tracking Service"

- from: "Commissioning Service"
to: "Workforce Management Service"

- from: "Commissioning Service"
to: "File Storage Service"

- from: "Commissioning Service"
to: "Notification Service"

- from: "Reporting Service"
to: "Workforce Management Service"

- from: "Reporting Service"
to: "Project and Work Tracking Service"

- from: "Reporting Service"
to: "CRM and Proposals Service"

- from: "Reporting Service"

to: "Financial Management Service"

- from: "Reporting Service"

to: "Billing Service"

- from: "Reporting Service"

to: "Commissioning Service"

- from: "Notification Service"

to: "Administration and Configuration Service"

- from: "Notification Service"

to: "Workforce Management Service"

- from: "Notification Service"

to: "Project and Work Tracking Service"

- from: "Notification Service"

to: "Commissioning Service"

- from: "File Storage Service"

to: "Administration and Configuration Service"

- from: "File Storage Service"

to: "Project and Work Tracking Service"

- from: "File Storage Service"

to: "Billing Service"

- from: "File Storage Service"

to: "Content and Document Management Service"

- from: "File Storage Service"

to: "Commissioning Service"